

SUSTAINABLE LOGISTICS ANALYTICS

Sustainable Logistics Optimization

Carbon Footprint & Cost Analysis

Leveraging data for ESG performance and operational efficiency



ESG & Business Objectives

Analyze, predict, and optimize logistics operations to reduce carbon footprint and improve cost efficiency — directly supporting corporate ESG goals.

KEY ESG FOCUS AREAS



Environmental

Greenhouse gas emissions reduction (CO₂e) and sustainable resource utilization



Operational

Enhanced supply chain efficiency and fuel consumption management

EXPECTED BUSINESS VALUE

- Improved brand reputation and stakeholder trust
- Compliance with evolving environmental regulations
- Realized cost savings through optimized fuel use and operational planning
- Competitive advantage in a sustainability-conscious market

Data Acquisition & Feature Engineering

Internal logistics operational data (dummy CSV) — 10 shipments across 19 analyzed metrics.

10

Shipments (rows)

19

Analyzed metrics
(columns)

3

Transit modes compared

ORIGINAL KEY FEATURES

- Shipment_ID, Date, Facility_Type, Urban_Zone
- Transit_Mode — primary determinant of emissions (Diesel Truck, Electric Van, Hybrid Van)
- Distance_km — distance traveled per shipment
- Medical_Payload_kg — weight of goods transported
- Traffic_Density_Index — environmental efficiency factor
- Actual_Fuel_Liters, Actual_CO2e_kg, Late_Delivery_Flag

ENGINEERED & CALCULATED FEATURES

- Predicted_CO2e_kg — model-generated CO2e emissions
- Emissions_Intensity_CO2e_per_kg — CO2e emitted per kg of payload
- Carbon_Savings_kg — reduction vs. all-diesel-truck baseline
- Actual_Fuel_Cost, Baseline_Fuel_Cost, Estimated_Fuel_Cost_Variance — financial impact metrics

Predictive Modeling

Goal: forecast Actual_CO2e_kg from shipment characteristics using Multiple Linear Regression (Scikit-Learn).

INDEPENDENT VARIABLES (PREDICTORS)

- Distance_km
- Medical_Payload_kg
- Traffic_Density_Index
- Transit_Mode (one-hot encoded)

Output: model generates Predicted_CO2e_kg per shipment, enabling proactive emissions monitoring.

MODEL PERFORMANCE (TEST SET)

R^2

~0.38

of variance in actual CO2e explained by the model

MSE

~2.12

avg. squared difference: predicted vs. actual CO2e

Pipeline note

Performance reflects a 10-shipment pilot dataset intended to validate the end-to-end pipeline. Accuracy is expected to improve materially as historical shipment volume grows.

ESG Reporting: Key Metrics

92.49 kg

Total Predicted Emissions (CO2e)

95.40 kg

Total Actual Emissions (CO2e)

0.0151

Mean Emissions Intensity (kg CO2e / kg payload)

31.14 kg

Total Carbon Savings vs. Baseline

0.0299

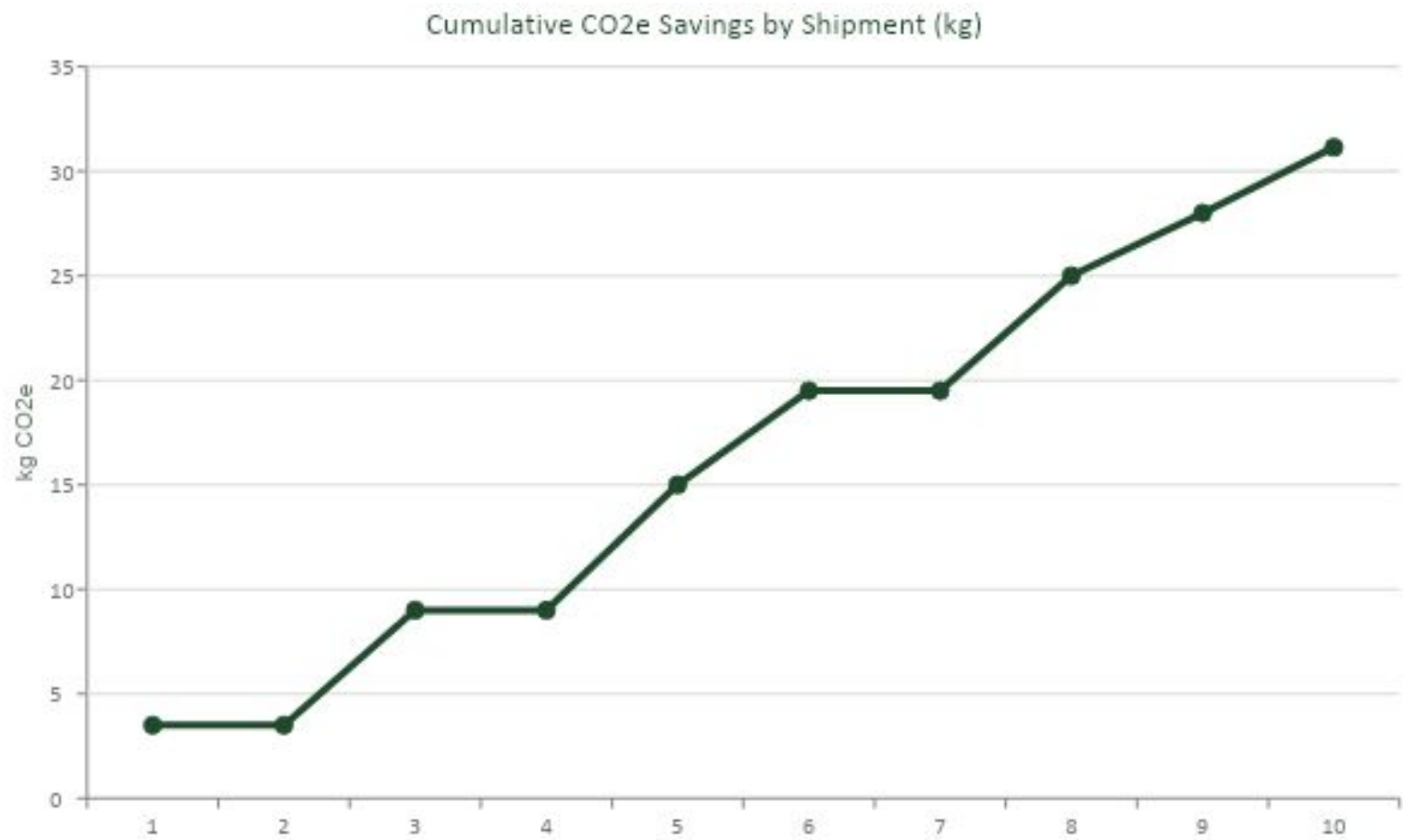
Late Delivery vs. High Emission Correlation

-\$15.75

Total Estimated Fuel Cost Variance

Late-delivery correlation is very weak — emissions are not a primary driver of delivery timeliness in this dataset.

Cumulative Carbon Savings Over Time



FINDING

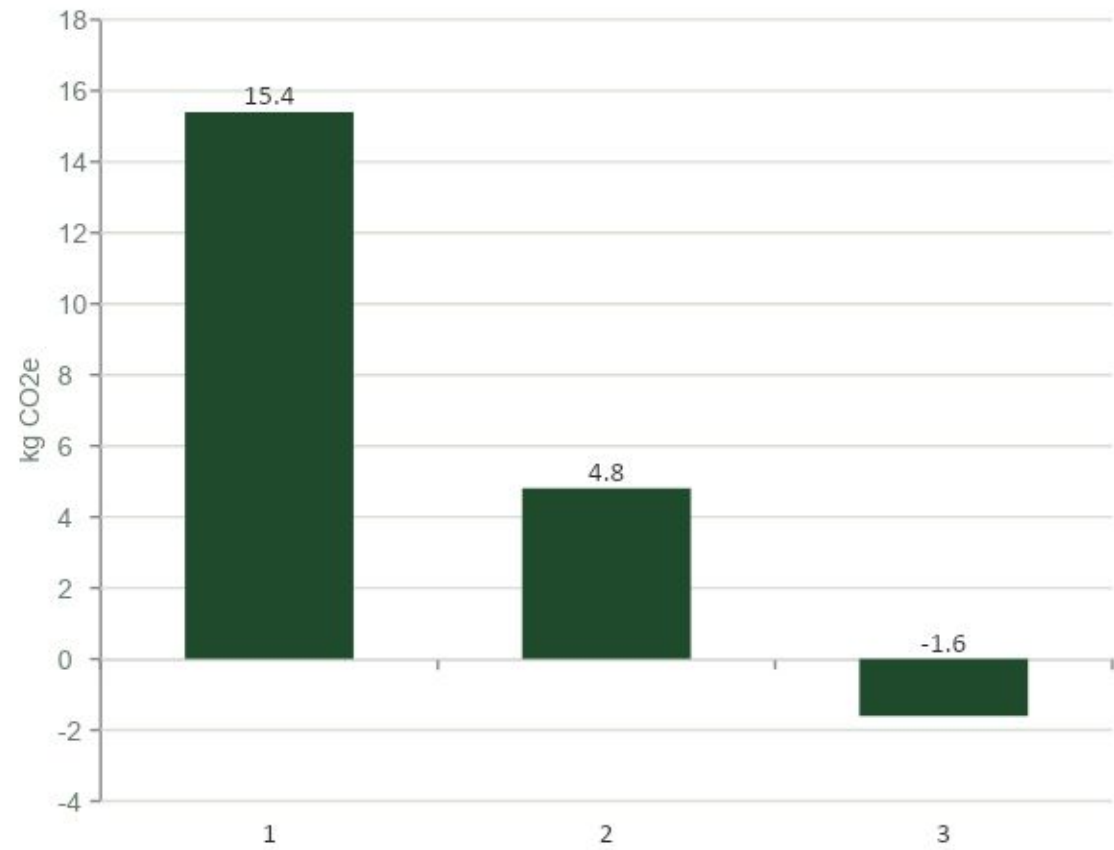
Cumulative carbon savings show a positive, growing trend — demonstrating effective emissions reduction over time, reaching over

31 kg CO2e

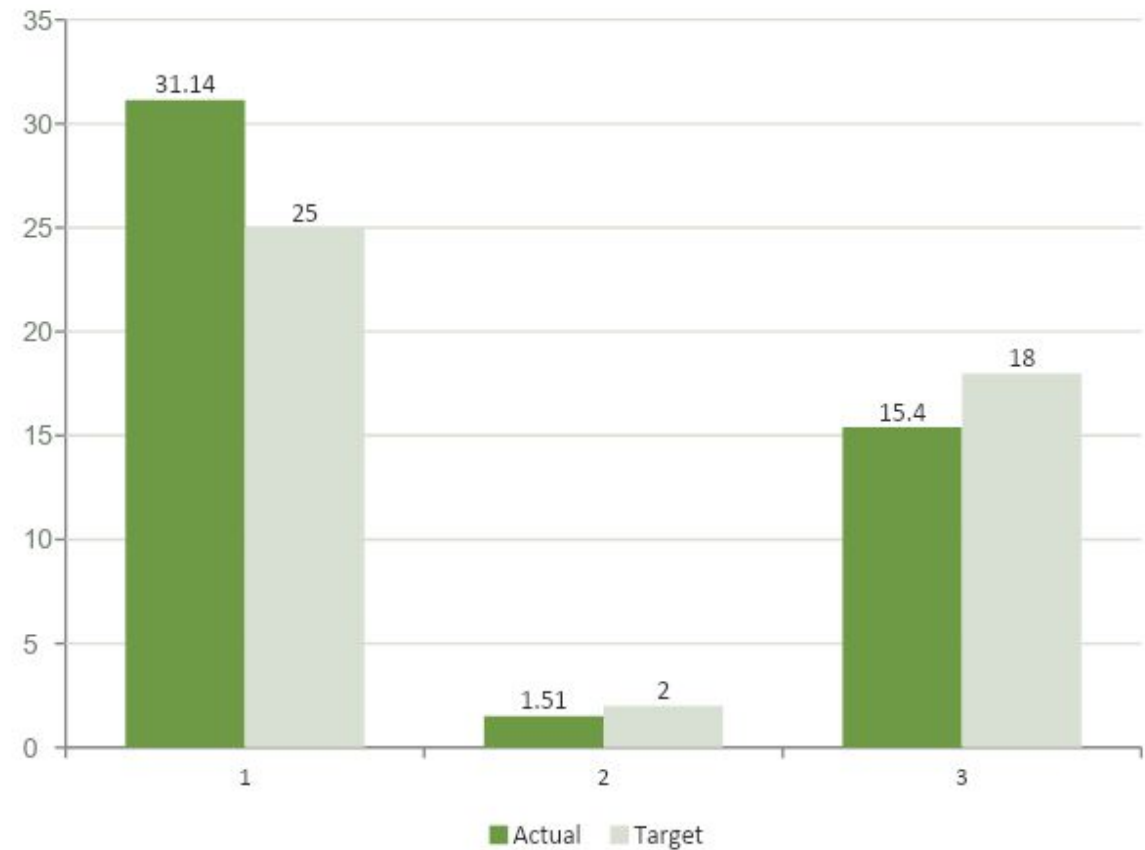
in total savings across the 10-shipment pilot.

Emissions by Transit Mode & ESG Target Attainment

AVG. PREDICTED CO2e EMISSIONS BY SHIPPING MODE



SUSTAINABILITY PERFORMANCE VS. CORPORATE ESG TARGETS



Summary Findings



Achieved Carbon Reductions

Quantifiable carbon savings through strategic logistics choices.



Impact of Transit Mode

Vehicle type is the most critical factor influencing CO2e — Electric and Hybrid Vans offer significant advantages.



Cost Efficiency

Operational changes produced a positive fuel cost variance — economic benefit alongside environmental gain.



ESG Target Attainment

Current performance exceeds all set ESG targets, showcasing strong sustainability progress.



Late Delivery Analysis

Very weak correlation between higher emissions and late deliveries — emissions focus alone won't move delivery timeliness.

Recommendations & Next Steps

1

Strategic Transit Mode Optimization

Systematically prioritize Electric & Hybrid Vans for feasible routes, especially high-density urban areas and lighter payloads.

2

Maximize Payload Utilization

Implement advanced load planning and route optimization tools to keep vehicles operating at optimal capacity.

3

Continuous ESG Target Setting

Given current success, revise and set more ambitious targets for emissions, fuel efficiency, and sustainable practices.

4

Leverage Advanced Analytics

Integrate real-time data for dynamic route adjustments, predictive maintenance, and sharper emissions forecasting.

5

Proactive Stakeholder Engagement

Regularly communicate sustainability performance, achievements, and future goals internally and externally.